

Package ‘flowMagic’

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Version 0.99

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Examples

```
add_labels_column()
```

```
assign_events_to_nearest_centroids  
    assign_events_to_nearest_centroids
```

Description

function to assign events to class with nearest centroid.

Usage

```
assign_events_to_nearest_centroids(  
    gated_df,  
    n_cores = 1,  
    method_dist = "euclidean",  
    thr_dist = 0.15,  
    include_zero = F,  
    remove_centroids = T  
)
```

Arguments

<code>gated_df</code>	dataframe with labels (third column).
<code>n_cores</code>	Number of cores. Default to 1.
<code>method_dist</code>	Distance method calculation. Default to euclidean.
<code>thr_dist</code>	Distance threshold for centroids calculation. Default to 0.15.
<code>include_zero</code>	Consider centroid of label 0. Default to False.
<code>remove_centroids</code>	Remove centroids too near each other based on thr_dist value.

Value

Dataframe.

Examples

```
assign_events_to_nearest_centroids()
```

check_polygons_intersection	
	<i>extract_polygon_gates</i>

Description

function to check polygons intersection.

Usage

check_polygons_intersection(list_df_hull)

Arguments

list_df_hull List of polygons coordinates

Value

float

Examples

check_polygons_intersection()

compute_gates	<i>compute_gates</i>
---------------	----------------------

Description

function to assign events based on polygon gates.

Usage

compute_gates(gated_df, list_final_polygons_coords, no_classes = F)

Arguments

- gated_df dataframe with labels (third column).
- list_final_polygons_coords
 List of dataframes containing polygon coordinates.
- no_classes Generate third column of labels. Default to False.

Value

Dataframe.

Examples

```
compute_gates()
```

```
convert_to_integers_chr
      convert_to_integers_chr
```

Description

function to text classes to integers-like classes if necessary.

Usage

```
convert_to_integers_chr(df)
```

Arguments

df Dataframe with classes (third column) to check and eventually modify

Value

Dataframe

Examples

```
convert_to_integers_chr()
```

```
csv_to_dens                      csv_to_dens
```

Description

function to get density of events (with classes associated if present).

Usage

```
csv_to_dens(df, with_classes = T, n_coord = "df", normalize_data = T)
```

Arguments

df Dataframe of marker expression values.
with_classes Consider classes. Default to True.
n_coord Grid size. Default to df.
normalize_data If True, data is normalized to 0-1 range. Default to True.

Value

Dataframe.

Examples

```
csv_to_dens()
```

exports_plots	<i>exports_plots</i>
---------------	----------------------

Description

function to generate plots (no hierarchy) from list of labelled dataframes.

Usage

```
exports_plots(  
  list_gated_data,  
  path_output,  
  n_cores = 1,  
  type_plot = "dens",  
  show_legend = T,  
  x_lab = "x",  
  y_lab = "y",  
  size_title_x = 23,  
  size_title_y = 23,  
  aspect_ratio = NULL,  
  w_val = 16,  
  h_val = 10,  
  size_axis_text = 25,  
  export_csv = F,  
  side_by_side = F,  
  ...  
)
```

Arguments

list_gated_data	list of dataframes. Each dataframe has 3 columns: marker 1 values, marker 2 values and label column.
path_output	Path to the directory where to export the plots.
n_cores	Number of cores to use. Default to 1.
type_plot	the user can choose between density (= "dens") or label assignment visualization (= "ML")
show_legend	If True it shows the legend for the label assignment visualization. Default to True.

<code>x_lab</code>	x-axis label.
<code>y_lab</code>	y-axis label.
<code>size_title_x</code>	Size x axis label.
<code>size_title_y</code>	Size y axis label.
<code>aspect_ratio</code>	Set aspect ratio. Default to NULL> If = 1, y and x axis ticks have same distance.
<code>w_val</code>	width value. Default to 16.
<code>h_val</code>	height value. Default to 10.
<code>size_axis_text</code>	Size of ticks labels.
<code>export_csv</code>	Export plot data as csv files. Default to False.
<code>side_by_side</code>	Arrange dens plot and ML plot side-by-side. Default to False.

Examples

```
exports_plots()
```

`export_raw_gs_plots` *export_raw_gs_plots*

Description

function to generate ungated plots from selected gs node

Usage

```
export_raw_gs_plots(
  gs,
  node_name,
  channel_x,
  channel_y,
  path_output,
  n_cores = 1,
  x_lab = NULL,
  y_lab = NULL,
  w_val = 7,
  h_val = 7,
  size_points = 1,
  return_data = F,
  ...
)
```


Arguments

<code>gs</code>	GatingSet
<code>path_output</code>	Path to the directory where to export the plots.
<code>n_cores</code>	Number of cores to use. Default to 1.
<code>x_lab</code>	x-axis label. Default to NULL (inherited from GatingSet metadata).
<code>y_lab</code>	y-axis label. Default to NULL (inherited from GatingSet metadata).
<code>w_val</code>	width value. Default to 7 inches.
<code>h_val</code>	height value. Default to 7 inches.
<code>size_points</code>	Size points scatter plot.
<code>return_data</code>	If TRUE, return the list of dataframes used to generate the plots. Default to FALSE.

Examples

```
export_raw_gs_plots()
```

```
extract_polygon_gates extract_polygon_gates
```

Description

function to extract the polygon gates objects based on the convex hull and classes.

Usage

```
extract_polygon_gates(gated_df, concavity_val = 1, ...)
```

Arguments

<code>gated_df</code>	dataframe with labels (third column).
<code>concavity_val</code>	Concavity of polygons. Default to 1.

Value

List of dataframes.

Examples

```
extract_polygon_gates()
```

```
flowmagic_pred_to_gs  flowmagic_pred_to_gs
```

Description

function to add list of polygon objects generated by flowmagic_pred_to_poly_gates function to GatingSet object.

Usage

```
flowmagic_pred_to_gs(list_poly_gates, gs, parent_node)
```

Arguments

list_poly_gates	List of polygonGate objects generated using the flowmagic_pred_to_poly_gates
gs	GatingSet to update with the flowMagic polygons.
parent_node	Parent population name of the new gate.

Value

GatingSet

Examples

```
flowmagic_pred_to_gs()
```

```
flowmagic_pred_to_poly_gates
flowmagic_pred_to_poly_gates
```

Description

function to label points of dataframe based on polygon coordinates

Usage

```
flowmagic_pred_to_poly_gates(
  list_df,
  pred_label,
  gate_label,
  n_cores = 1,
  concavity_val = 10
)
```

Arguments

list_df	List of dataframes with three columns: marker 1, marker 2, class. Can get input generated by the magicPred function.
pred_label	Select prediction label to extract polygon.
gate_label	Label for the selected polygon within the GatingSet framework.
n_cores	Number of cores to use. Default to 1.
concavity_val	Concavity value for drawing polygon. Default to 10.

Value

list

Examples

```
flowmagic_pred_to_poly_gates()
```

get_centroids	<i>get_centroids</i>
---------------	----------------------

Description

function to get centroids for each label

Usage

```
get_centroids(
  df,
  low_thr = 0.1,
  up_thr = 0.9,
  thr_dist = 0.15,
  include_zero = F,
  remove_centroids = T
)
```

Arguments

df	dataframe with labels (third column).
low_thr	Lower threshold for quantile calculation.
up_thr	Upper threshold for quantile calculation.
thr_dist	Distance threshold for centroids calculation. Default to 0.15.
include_zero	Consider centroid of label 0. Default to False.
remove_centroids	Remove centroids too near each other based on thr_dist value.

Value

Dataframe.

Examples

```
get_centroids()
```

```
get_classes_expr_df     get_classes_expr_df
```

Description

function to get classes of original expression df based on density df predictions.

Usage

```
get_classes_expr_df(dens_df, original_df)
```

Arguments

<code>dens_df</code>	Dataframe of density values.
<code>original_df</code>	Original dataframe.

Value

Dataframe.

Examples

```
get_classes_expr_df()
```

```
get_density_features     get_density_features
```

Description

function to get density features only given a bivarite density csv.

Usage

```
get_density_features(df_dens, min_height = 0.06)
```

Arguments

<code>df_dens</code>	Dataframe of density estimates for both markers.
<code>min_height</code>	Minimum height of the peaks to consider.

Value

Vector of numbers.

Examples

```
get_density_features()
```

get_density_scores	<i>get_density_scores</i>
--------------------	---------------------------

Description

function to get scores for distance template calculation.

Usage

```
get_density_scores(df_template, df_test, select_density_features = NULL)
```

Arguments

df_template	Dataframe of template markers expression values.
df_test	Dataframe of test markers expression values.
select_density_features	Select features to use. Default to NULL.

Value

Matrix of numbers.

Examples

```
get_density_scores()
```

get_distance_loc_vs_test	<i>get_distance_loc_vs_test</i>
--------------------------	---------------------------------

Description

function to compare the similiraty between current test data and local training set.

Usage

```
get_distance_loc_vs_test(test_df, loc_df, show_plot = "none", nboot = 50)
```

Arguments

<code>test_df</code>	Dataframe with bivariate marker expression.
<code>loc_df</code>	Dataframe with bivariate marker expression.
<code>show_plot</code>	show density comparison. Default to none.
<code>nboot</code>	Number of permutations. Default to 50.

Value

List of p-values.

Examples

```
get_distance_loc_vs_test()
```

<code>get_dist_template</code>	<i>get_dist_template</i>
--------------------------------	--------------------------

Description

function to get distance between template and test data.

Usage

```
get_dist_template(matrix_scores, dist_method = "euclidean")
```

Arguments

<code>matrix_scores</code>	Matrix of density features generated by <code>get_density_scores</code> function.
<code>dist_method</code>	Type of distance method calculation.

Value

Number.

Examples

```
get_dist_template()
```

```
get_flowframe_from_gs  get_flowframe_from_gs
```

Description

function to get flowframe from gs (converting cytoframe to flowframe)

Usage

```
get_flowframe_from_gs(gs, node_name, sample_id)
```

Arguments

gs	GatingSet
node_name	Name of the Gating tree node whose gating data needs to be extracted.
sample_id	Name or index of the sample to extract.

Value

List.

Examples

```
get_flowframe_from_gs()
```

```
get_hierarchy_all_pops
      get_hierarchy_all_pops
```

Description

function to get the hierarchy of all pops from the sample manually gated (the input gating hierarchy). Based on its output the function magicTrain will perform the training step using the sample manually gated (local training set) and the project discovery data (global training set).

Usage

```
get_hierarchy_all_pops(gh, export_visnet = F, path.output = "None")
```

Arguments

gh	GatingHierarchy.
export_visnet	If true, it export visnetwork object.
path.output	Path to save visnetwork object. Default to None.

Value

List of Dataframes.

Examples

```
get_hierarchy_all_pops()
```

<code>get_hull_all_gates</code>	<i>get_hull_all_gates</i>
---------------------------------	---------------------------

Description

function to get the convex hull of all gates.

Usage

```
get_hull_all_gates(gated_df, concavity_val = 1, spar_val = 0.7, smoothing = F)
```

Arguments

- `gated_df` dataframe with labels (third column).
- `concavity_val` Values of concavity. Default to 1.
- `spar_val` Value of spar. Default to 0.7.
- `smoothing` Apply smoothing of lines? Default to F.

Value

List of dataframes.

Examples

```
get_hull_all_gates()
```

`get_indices_cross_val` *get_indices_cross_val*

Description

function to get indices for cross val.

Usage

```
get_indices_cross_val(  
    df_train,  
    n_cores = 1,  
    train_inds = "plot_num",  
    val_inds = "none",  
    n_train_plots = 5,  
    n_folds = 5,  
    seed = 40,  
    n_val_plots = 5  
)
```

Arguments

<code>df_train</code>	Dataframe of training features generated by the <code>get_train_data</code> function.
<code>n_cores</code>	Number of cores to use. Default to 1.
<code>train_inds</code>	Type of method to extract training indices: <code>plot_num</code> , <code>rand_set_num</code> , <code>rand_set_n_gates_info</code> .
<code>val_inds</code>	Type of method to extract validation indices: <code>plot_num</code> , <code>rand_set_num</code> , <code>rand_set_n_gates_info</code> .
<code>n_train_plots</code>	Number of training plots to include in training data for each iteration.
<code>n_folds</code>	Number of training iterations.
<code>seed</code>	Seed to randomly extract data.
<code>n_val_plots</code>	Number of training plots to include in validation data for each iteration.

Value

List of integers.

Examples

```
get_indices_cross_val()
```

```
get_list_df_gated_plots
      get_list_df_gated_plots
```

Description

function to get labelled dataframe based on selected node for each sample in gs.

Usage

```
get_list_df_gated_plots(gs, gate_name, label_pop = NULL)
```

Arguments

gs	GatingSet
gate_name	Name of the Gating tree node whose gating data needs to be extracted.
label_pop	Set label for selected gate.

Value

List.

Examples

```
get_list_df_gated_plots()
```

```
get_local_train_sets  get_local_train_sets
```

Description

Based on the hierarchy calculated using the get_hierarchy function, we generate the local training sets gated by the biologists.

Usage

```
get_local_train_sets(gh, hierarchical_tree, info_hierarchy)
```

Arguments

gh	GatingHierarchy.
hierarchical_tree	Dataframe of hierarchy information generated by the get_hierarchy function.
info_hierarchy	List of hierarchy information generated by the get_hierarchy function.

Value

List of Dataframes.

Examples

```
get_local_train_sets()
```

```
get_pops_hierarchy_list  
get_pops_hierarchy_list
```

Description

function to get info from a list of hierarchical dataset (it reports all the pops for each level in a vector).

Usage

```
get_pops_hierarchy_list(hierarchical_list)
```

Arguments

`hierarchical_list`
list of populations names for each level.

Value

Vector of characters.

Examples

```
get_pops_hierarchy_list()
```

```
get_pop_multiclass    get_pop_multiclass
```

Description

function to get the children of the selected population, useful for multiclass classification. The output is only the pops with same dimensions

Usage

```
get_pop_multiclass(gh, pop)
```

Arguments

gh	GatingHierarchy.
pop	Name of population to get events assignments.

Value

Vector of characters.

Examples

```
get_pop_multiclass()
```

get_slot_hierarchy_list
<i>get_slot_hierarchy_list</i>

Description

function to access a slot of results from the hierarchy list based on the pop selected.

Usage

```
get_slot_hierarchy_list(hierarchical_list, pop_selected)
```

Arguments

hierarchical_list	list of populations names for each level.
pop_selected	Population name.

Value

Element of a list.

Examples

```
get_slot_hierarchy_list()
```

get_test_sets	<i>get_test_sets</i>
---------------	----------------------

Description

Function to obtain the data ready to be gated (validation set or test set). It takes as input a flowSet and generates a list of dataframes with a ML structure for each ungated fcs file. Each dataframe is the expression matrix of the root.

Usage

```
get_test_sets(fs, gh)
```

Arguments

fs	flowSet to gate.
gh	Gating hierarchy.

Value

List of dataframes.

Examples

```
get_test_sets()
```

get_train_data	<i>get_train_data</i>
----------------	-----------------------

Description

function to import training data based on paths to files.

Usage

```
get_train_data(  
  paths_file = NULL,  
  df_paths = NULL,  
  n_cores = 1,  
  prop_down = NULL,  
  remove_class = NULL,  
  n_points_per_plot = NULL,  
  normalize_data = T,  
  vec_col = NULL  
)
```

Arguments

<code>paths_file</code>	Vector of paths. Each path points toward a single csv file containin training info (labels and bivariate expression). <code>paths_file</code> can be also directly the list of dataframes containing labels and bivariate expression.
<code>df_paths</code>	Dataframe containing the paths of file to read. The paths to data must be in the first column. The associated paths to classes are in second column.
<code>n_cores</code>	Number of cores. Default to 1.
<code>prop_down</code>	Proportion of events (downsampling). Default to NULL (downsampling using number of points).
<code>remove_class</code>	Vector of classes to ignore. Default to NULL.
<code>n_points_per_plot</code>	Number of points for downsampling.
<code>normalize_data</code>	If True, data is normalized to 0-1 range. Default to True.
<code>vec_col</code>	vector of columns names if the input dataframes have more than 3 columns. The third column name must always refer to the column with the gate label of each event. Default to NULL.

Value

Dataframe.

Examples

```
get_train_data()
```

```
get_weights_density_features
      get_weights_density_features
```

Description

function to calculate final score based on density features.

Usage

```
get_weights_density_features(df_scores)
```

Arguments

<code>df_scores</code>	Dataframe of distance scores.
------------------------	-------------------------------

Value

Vector of numbers.

Examples

```
get_weights_density_features()
```

import_gating_info	<i>import_sample_gated</i>
--------------------	----------------------------

Description

convert a flowWorkspace or a GatingML file of the train data folder into a gated gh (gh_train)

Usage

```
import_gating_info(path, type = "gs", group_wsp = NULL)
```

Arguments

path	path to gs or GatingML object.
type	type of object.
group_wsp	Group of wsp to import.

Value

GatingSet object

Examples

```
import_sample_gated()
```

import_reference_csv	<i>import_reference_csv</i>
----------------------	-----------------------------

Description

function to import plain gold standards data (no hierarchy)

Usage

```
import_reference_csv(path_results, n_cores = 1)
```

Arguments

path_results	path to directory containing the csv files to read (with third column of labels).
n_cores	Number of cores to use. Default to 1.

Value

list of dataframes

Examples

```
import_reference_csv()
```

```
import_test_set_csv    import_test_set_csv
```

Description

function to import test set in csv format.

Usage

```
import_test_set_csv(path_data, n_cores = 1, xy_col = T)
```

Arguments

path_data	path to directory containing csv files to read (third column is ignored).
n_cores	Number cores. Default to 1.
xy_col	Colnames equal to x and y. Default to True.

Value

List of dataframes.

Examples

```
import_test_set_csv()
```

```
import_test_set_fcs    import_test_set
```

Description

Read the ungated fcs files into a flowSet.

Usage

```
import_test_set_fcs(
  path,
  n_samples = "All",
  ref_f_n = 1,
  pattern_filter = NULL
)
```

Arguments

path	path of directory containig the fcs files.
n_samples	Number of samples. Default to All.
ref_f_n	Set reference flowFrame to match channel names. Default to 1(first flowFrame).
pattern_filter	Filter files to import based on string patterns. Default to NULL.

Value

flowSet.

Examples

```
import_test_set()
```

magicGating

magicGating

Description

function to manually gate samples in a flowSet.

Usage

```
magicGating(
  fs,
  sample_id = 1,
  channel_x,
  channel_y,
  gs_node = NULL,
  label_pol = "1",
  ...
)
```

Arguments

fs	An object of class flowSet. Can be also an object of class cytoSet (it will be converted to flowSet) or a GatingSet object (in this case, gs_node is mandatory)
sample_id	Names of the samples to gate. It can also be the numerical index of the sample. Default to sample 1.
channel_x	Name of the channel (x-axis).
channel_y	Name of the channel (y-axis).
gs_node	Name of the node to extract data, if fs is a GatingSet object this a mandatory argument.
label_pol	Label of the gate polygon. Default to "1".

Value

List of two objects of class List. list_poly_gates: List of polygon coordinates. Each element refers to the coordinates of one sample. list_gated_data: List of gated data. Each element refers to the gated data of one sample.

Examples

```
magicGating()
```

magicPlot

magicPlot

Description

function to generate the scatter plot with colored density of the events.

Usage

```
magicPlot(
  df,
  type = "dens",
  polygons_coords_list = NULL,
  show_legend = T,
  size_axis_text = 18,
  size_title_x = 20,
  size_title_y = 20,
  treat_0_as_gate = F,
  x_lab = NULL,
  y_lab = NULL,
  gates_to_plot = NULL,
  apply_manual_scale = F,
  hull_only = F,
  size_points = 1,
  concavity_val = 20,
  aspect_ratio = NULL,
  x_lim1 = NULL,
  x_lim2 = NULL,
  y_lim1 = NULL,
  y_lim2 = NULL,
  add_labels = F,
  map_label_polygon = NULL,
  size_pol_name = 6,
  show_marginals = F,
  ...
)
```

Arguments

df	Dataframe of bivariate markers expression (with labels if gates to plot).
type	Type of plot to generated. "dens"= plot with bivariate density. "ML"= plot with color layer based on machine learning class assignments
polygons_coords_list	list of gates coordinates. Needed if labels not included in df. Default to NULL.
show_legend	Show legend if type="ML". Default to True.
size_axis_text	Size of axis ticks labels. Default to 18.

size_title_x	Size of x axis label title. Default to 20.
size_title_y	Size of y axis label title. Default to 20.
treat_0_as_gate	Treat 0 label as gate. Default to False (0 label is background)
x_lab	Label of x axis. Default to NULL (inherited from input dataframe by default).
y_lab	Label of y axis. Default to NULL (inherited from input dataframe by default).
gates_to_plot	Select labels to plot.
apply_manual_scale	Apply predefined scale of colors. Default to True.
size_points	Size of points in scatter plot.
concavity_val	Concavity value. Default to 20. Higher value, less jagged boundaries
aspect_ratio	Aspect ratio value. If = 1, y and x axis have equal distance between the ticks labels. Default to NULL.
x_lim1	Minimum limit x axis. Default to NULL.
x_lim2	Max limit x axis. Default to NULL.
y_lim1	Minimum limit y axis. Default to NULL.
y_lim2	Max limit y axis. Default to NULL.
add_labels	add polygon labels. Default to FALSE.
map_label_polygon	map of polygon labels (to assign custom labels). Default to NULL.
size_pol_name	size polygon labels. Default to 6.
show_marginals	show 1D density next to axis. Default to False.

Value

Object of class ggplot

Examples

```
magicPlot()
```

magicplot_3D

magicplot_3D

Description

function to make a 3D plot of data.

Usage

```
magicplot_3D(  
  df,  
  class_col = F,  
  x_lab = "x",  
  y_lab = "y",  
  z_lab = "z",  
  type = "ML",  
  size_p = 1  
)
```

Arguments

df	input dataframe composed of at least three columns. Four columns if gate needs to be plot.
class_col	Gate to plot? Default to False.
x_lab	label of the x axis
y_lab	Label of the y axis
z_lab	Label of the z axis
type	Assuming class_col==T, If type==ML, Generate a plot colored based on the gate assignment. If type=="mesh", generate a 3d polygon gate.
size_p	size of scatter plot points. Default to 1.

Value

plotly plot

Examples

```
magicplot_3D()
```

<code>magicPlot_fs</code>	<i>magicPlot_fs</i>
---------------------------	---------------------

Description

function to plot FCM data directly from objects of class flowSet or flowFrame. The data is shown in a density scatter plot.

Usage

```
magicPlot_fs(fs, sample_id, channel_x, channel_y, ...)
```

Arguments

fs	object of class flowSet or flowFrame.
sample_id	Name of the sample data to plot. No effect if fs is an object of class flowFrame.
channel_x	Name of the channel to plot (x-axis).
channel_y	Name of the channel to plot (y-axis).

Value

Object of class ggplot

Examples

```
magicPlot_fs()
```

magicPlot_template	<i>magicPlot_template</i>
--------------------	---------------------------

Description

function to interactively generate a template based on bivariate ungated data.

Usage

```
magicPlot_template(df, size_points = 1)
```

Arguments

df	input dataframe composed of two columns reporting markers expression (first column = marker 1, second column = marker 2).
size_points	Size of points in scatter plot. Default to 1.

Value

Dataframe

Examples

```
magicPlot_template()
```

magicPred

magicPred

Description

function to predict on plain test data (no hierarchy)

Usage

```
magicPred(
  test_data,
  magic_model = NULL,
  magic_model_n_gates = NULL,
  ref_model_info = NULL,
  n_cores = 1,
  ref_data_train = NULL,
  prop_down = NULL,
  thr_dist = 0.05,
  n_points_per_plot = NULL,
  normalize_data = T,
  include_zero_val = T,
  ...
)
```

Arguments

test_data	Dataframe of test data to gate. It has only the two columns of marker expression.
magic_model	Global trained model to predict gates. It can be a single model or list of named models (each model trained on selected number of gates).
magic_model_n_gates	Global trained model to predict number of gates. If different from NULL, magic_model is expected to be a list of models to predict certain gates (e.g., 5 models for 2,3,4,5 or 6 gates).
ref_model_info	Template model to predic gates.
n_cores	Number of cores to use. Default to 1.
ref_data_train	Template data used to generate ref_model_info. Needed to calculate target-template distance.
prop_down	Proportion for downsampling. Default to NULL (automatic downsampling using n_points_per_plot).
n_points_per_plot	Number of points to consider for downsampling. Default to 500.
normalize_data	If True, data is normalized to 0-1 range. Default to True.
include_zero_val	considering events labeled as 0 as an additional gate when there is only one gate. Default to True.

Value

List of Dataframes.

Examples

magicPred()

magicPred_all	<i>magicPred_all</i>
---------------	----------------------

Description

function to predict on plain test data (no hierarchy)

Usage

```
magicPred_all(  
  list_test_data,  
  sample_id = NULL,  
  magic_model = NULL,  
  ref_model_info = NULL,  
  magic_model_n_gates = NULL,  
  ref_data_train = NULL,  
  prop_down = NULL,  
  n_points_per_plot = NULL,  
  n_gates_df = NULL,  
  thr_dist = 0.05,  
  n_cores = 1,  
  normalize_data = T,  
  include_zero_val = T,  
  n_cores_all = 1,  
  verbose = F,  
  ...  
)
```

Arguments

- list_test_data List of unlabeled dataframes. It has only the two columns of marker expression.
- sample_id Id of the samples to gate. Could be a vector containing the names or numerical indices (position) of the samples to gate. If NULL, all samples in list_test_data are gated. Default to NULL.
- magic_model Global trained model to predict gates. It can be a single model or list of named models (each model trained on selected number of gates).
- ref_model_info Template model to predic gates.

magic_model_n_gates	Global trained model to predict number of gates. If different from NULL, magic_model is expected to be a list of models to predict certain gates (e.g., 5 models for 2,3,4,5 or 6 gates).
ref_data_train	Template data used to generate ref_model_info. Needed to calculate target-template distance.
n_points_per_plot	Number of points to consider for downsampling. Default to 500. Note: the gates in the downsampled data are projected back to the original data to get the actual populations.
n_gates_df	Dataframe pairing the sample name (first column) with the user-defined number of gates (second column). Default to NULL.
thr_dist	Distance threshold for centroids calculation during post processing. Default to 0.15.
n_cores	Number of cores to use to process one sample. Default to 1.
normalize_data	If True, data is normalized to 0-1 range. Default to True.
include_zero_val	considering events labeled as 0 as an additional gate when there is only one gate. Default to True.
n_cores_all	Number of cores to use across all samples. Default to 1.
verbose	If True, print all message and disable tryCatch (any error will stop the execution). Default to False.

Value

List of Dataframes.

Examples

```
magicPred_all()
```

magicPred_hierarchy	<i>magicPred_hierarchy</i>
---------------------	----------------------------

Description

function to predict the gates on the ungated .fcs samples.

Usage

```
magicPred_hierarchy(list_test_sets, list_models_local, df_tree, n_cores = 1)
```


Arguments

`list_test_sets` contains the list of root dataframe for each ungated fcs file imported.

`list_models_local` contains the optimized local models pre-generated using the `magicTrain_local` function.

`df_tree` contains the info related to the populations hierarchy.

`n_cores` Number of cores to use. Default to 1.

Value

List of Dataframes.

Examples

```
magicPred_hierarchy()
```

magicTrain

magicTrain

Description

function to generate one training model based on a list of training sets (no hierarchy).

Usage

```
magicTrain(
  df_train,
  n_cores = 1,
  train_model = "rf",
  k_cv = 10,
  list_index_train = NULL,
  list_index_val = NULL,
  n_tree = 10,
  tune_lenght = 3,
  size_nnet_units = 100,
  decay_nnet = 0.1,
  method_control = "auto",
  type_y = "classes",
  seed_n = 40
)
```

Arguments

df_train	training dataframe generated by the get_train_data function.
n_cores	Number of cores to use. Default to 1.
train_model	Type of training model. Default to rf.
k_cv	Number of k for cross-validation (if method control=cv)
list_index_train	List of vector of indices to use in training for each fold.
list_index_val	List of vector of indices to use as held out data for each fold.
n_tree	Number of trees for random forest.
tune_lenght	Number of parameters values trained during cross-validation.
size_nnet_units	Number of units in hidden layer (if train_model=nnet).
decay_nnet	Decay parameter value for nnet model.
method_control	Type of training control: oob or cv or auto. Default to auto.
type_y	Type of response variable: classes (train to predict gates boundaries) or n_gates_info(train to predict number of gates).
seed_n	Set seed. Default to 40.

Value

model object.

Examples

```
magicTrain()
```

magicTrain_dt

magicTrain_dt

Description

function to generate a random forest training model.

Usage

```
magicTrain_dt(
  Xtrain,
  Ytrain,
  k_cv = 10,
  list_index_train = NULL,
  list_index_val = NULL,
  tune_lenght = 5
)
```

Arguments

Xtrain	Dataframe of training features.
Ytrain	Dataframe of labels (one column).
k_cv	Number of k for cross-validation.
list_index_train	List of vector of indices to use in training for each fold.
list_index_val	List of vector of indices to use as held out data for each fold.
tune_lenght	Number of hyper parameters to test. Default to 5.

Value

model object.

Examples

```
magicTrain_dt()
```

```
magicTrain_hierarchy    magicTrain_local
```

Description

function to generate the local training models using the list of hierarchical training set.

Usage

```
magicTrain_hierarchy(
  list_train_sets,
  n_tree = 10,
  train_model = "rf",
  method_control = "oob",
  n_cores = 1
)
```

Arguments

list_train_sets	List of labeled dataframe to train generated by the get_local_train function.
n_tree	Number of tree for random forest model.
train_model	Type of training model. Default to random forest ("rf").
method_control	Cross-validation method. Default to out-of-the-bag method (oob).
n_cores	Number of cores to use. Default to 1.

Value

List of models objects.

Examples

```
magicTrain_local()
```

magicTrain_knn	<i>magicTrain_knn</i>
----------------	-----------------------

Description

function to generate a random forest training model.

Usage

```
magicTrain_knn(
  Xtrain,
  Ytrain,
  k_cv = 10,
  list_index_train = NULL,
  list_index_val = NULL,
  tune_lenght = 5
)
```

Arguments

Xtrain	Dataframe of training features.
Ytrain	Dataframe of labels (one column).
k_cv	Number of k for cross-validation.
list_index_train	List of vector of indices to use in training for each fold.
list_index_val	List of vector of indices to use as held out data for each fold.
tune_lenght	Number of hyper parameters to test. Default to 5.

Value

model object.

Examples

```
magicTrain_knn()
```

magicTrain_nb	<i>magicTrain_nb</i>
---------------	----------------------

Description

function to generate a random forest training model.

Usage

```
magicTrain_nb(  
  Xtrain,  
  Ytrain,  
  k_cv = 10,  
  list_index_train = NULL,  
  list_index_val = NULL,  
  tune_lenght = 5  
)
```

Arguments

Xtrain	Dataframe of training features.
Ytrain	Dataframe of labels (one column).
k_cv	Number of k for cross-validation.
list_index_train	List of vector of indices to use in training for each fold.
list_index_val	List of vector of indices to use as held out data for each fold.
tune_lenght	Number of hyper parameters to test. Default to 5.

Value

model object.

Examples

```
magicTrain_nb()
```

magicTrain_nnet	<i>magicTrain_nnet</i>
-----------------	------------------------

Description

function to generate a neural net training model.

Usage

```
magicTrain_nnet(  
  Xtrain,  
  Ytrain,  
  k_cv = 10,  
  list_index_train = NULL,  
  list_index_val = NULL,  
  size = 100,  
  decay = 0.1,  
  tune_lenght = 5  
)
```

Arguments

Xtrain	Dataframe of training features.
Ytrain	Dataframe of labels (one column).
k_cv	Number of k for cross-validation.
list_index_train	List of vector of indices to use in training for each fold.
list_index_val	List of vector of indices to use as held out data for each fold.
size	Number of units in hidden layer (if train_model=nnet).
decay	Decay parameter value for nnet model.

Value

model object.

Examples

```
magicTrain_nnet()
```

magicTrain_rf	<i>magicTrain_rf</i>
---------------	----------------------

Description

function to generate a random forest training model.

Usage

```
magicTrain_rf(  
  Xtrain,  
  Ytrain,  
  list_index_train = NULL,  
  list_index_val = NULL,  
  n_tree = 10,  
  method_control = "oob",  
  k_cv = 10  
)
```

Arguments

Xtrain	Dataframe of training features.
Ytrain	Dataframe of labels (one column).
list_index_train	List of vector of indices to use in training for each fold.
list_index_val	List of vector of indices to use as held out data for each fold.
n_tree	Number of trees for random forest.
method_control	Type of training control: oob or cv. Default to oob.
k_cv	Number of k for cross-validation (if method control=cv).

Value

model object.

Examples

```
magicTrain_rf()
```

<code>magic_label_poly</code>	<i>magic_label_poly</i>
-------------------------------	-------------------------

Description

function to label points of dataframe based on polygon coordinates

Usage

```
magic_label_poly(df, polygon_df, label_pol = "1")
```

Arguments

<code>df</code>	Dataframe composed of two columns for marker expression of first (x axis = first column) and second marker (y axis = second column)
<code>polygon_df</code>	Dataframe containing the x coordinates (first column) and y coordinates (second column) of the current polygon to label.
<code>label_pol</code>	Label for current polygon.

Value

Dataframe

Examples

```
magic_label_poly()
```

<code>magic_label_rectangle</code>	<i>magic_label_rectangle</i>
------------------------------------	------------------------------

Description

function to label points of dataframe based on rectangle coordinates

Usage

```
magic_label_rectangle(df, x_min, x_max, y_min, y_max, label_pol = "1")
```

Arguments

<code>df</code>	Dataframe composed of two columns for marker expression of first (x axis = first column) and second marker (y axis = second column)
<code>x_min</code>	x coordinate minimum
<code>x_max</code>	x coordinate maximum
<code>y_min</code>	y coordinate minimum
<code>y_max</code>	y coordinate maximum
<code>label_pol</code>	Label for current polygon.

Value

Dataframe

Examples

```
magic_label_rectangle()
```

```
magic_manual_gating    magic_manual_gating
```

Description

Perform interactive manual gating on selected samples and add or update the resulting polygon gate in a GatingSet.

Usage

```
magic_manual_gating(
  gs_input,
  parent_node,
  new_child_node,
  samples_id,
  channel_x,
  channel_y,
  size_points = 0.5,
  return_list_out = F,
  input_list_out = NULL,
  return_gs_only = T,
  ...
)
```

Arguments

<code>gs_input</code>	A GatingSet object.
<code>parent_node</code>	Character. Name of the parent node where the new gate should be added or updated.
<code>new_child_node</code>	Character. Name of the child node to create or update.
<code>samples_id</code>	Character vector. Sample names to manually gate. These names must match <code>sampleNames(gs_input)</code> .
<code>channel_x</code>	Character. Name of the channel to plot on the x-axis.
<code>channel_y</code>	Character. Name of the channel to plot on the y-axis.
<code>size_points</code>	Numeric. Point size used in the interactive gating plot. Default is 0.5.
<code>return_list_out</code>	Logical. If TRUE, return only the output from <code>magicGating()</code> without modifying the GatingSet. Default is FALSE.

`input_list_out` Optional list. Existing output from `magicGating()` containing a `list_poly_gates` slot. If provided, interactive gating is skipped and these polygon gates are added or updated in the `GatingSet`.

`return_gs_only` Logical. If `TRUE`, return only the modified `GatingSet`. If `FALSE`, return a list containing the modified `GatingSet` and the gating output. Default is `TRUE`.

`...` Additional arguments passed to `magicGating()`.

Details

This function wraps `magicGating()` for one manual polygon gate and integrates the resulting `polygonGate` into a `GatingSet`. If the target child node does not already exist, it is created under `parent_node`. If the node already exists, the gate is updated only for the samples present in the gating output.

Value

If `return_gs_only = TRUE`, a modified `GatingSet`. Otherwise, a list with two elements: `gs`, the modified `GatingSet`, and `list_out`, the gating output used to add or update the gate.

Examples

```
gs <- magic_manual_gating(
  gs_input = gs,
  parent_node = "root",
  new_child_node = "Live_cells",
  samples_id = c("sample_01.fcs"),
  channel_x = "FSC-A",
  channel_y = "LIVE DEAD Blue-A"
)
```

`magic_plot_gs_sample` *magic_plot_gs_sample*

Description

Plot one sample from a `GatingSet` with one or more gated populations overlaid.

Usage

```
magic_plot_gs_sample(
  gs,
  sample_id,
  gate_names,
  size_points = 0.5,
  concavity_val = 50,
  ...
)
```

Arguments

<code>gs</code>	A GatingSet object.
<code>sample_id</code>	Character. Name of the sample to plot. Must be present in <code>sampleNames(gs)</code> .
<code>gate_names</code>	Character vector. One or more GatingSet node names to extract and display.
<code>size_points</code>	Numeric. Point size passed to <code>magicPlot()</code> . Default is 0.5.
<code>concavity_val</code>	Numeric. Concavity parameter available for downstream polygon plotting. Default is 50.
<code>...</code>	Additional arguments passed to <code>magicPlot()</code> , such as <code>type</code> , <code>size_axis_text</code> , <code>size_title_x</code> , or <code>size_title_y</code> .

Details

This function extracts one or more gated populations from a single sample in a GatingSet and visualizes them using `magicPlot()`. If multiple gate names are supplied, the extracted gated data are merged so the selected populations are displayed together on the same bivariate plot. Each population is labeled using its node name.

Value

A ggplot object showing the selected sample and gated populations.

Examples

```
# Plot one gate
p1 <- magic_plot_gs_sample(
  gs = gs,
  sample_id = "sample_01.fcs",
  gate_names = "Live_cells",
  size_points = 0.5
)

# Plot multiple gates together
p2 <- magic_plot_gs_sample(
  gs = gs,
  sample_id = "sample_01.fcs",
  gate_names = c("Live_cells", "Dead_cells"),
  type = "ML",
  size_points = 0.5
)
```

<code>magic_plot_wrap</code>	<i>magic_plot_wrap</i>
------------------------------	------------------------

Description

function to wrap all plots in one plot

Usage

```
magic_plot_wrap(list_gated_data, n_col_wrap = 3, size_title = 10, ...)
```

Arguments

<code>list_gated_data</code>	List of dataframes composed by three columns (could be generated by the <code>get_list_df_gated_plots()</code> function)
<code>n_col_wrap</code>	number of columns in the wrapped plot. Default to 3.
<code>size_title</code>	size of title of each plot. Default to 10.

Value

wrapped plot

Examples

```
magic_plot_wrap()
```

<code>magic_template_predict</code>	<i>magic_template_predict</i>
-------------------------------------	-------------------------------

Description

Predict gated populations from a trained flowMagic template model and optionally add the predicted gates to a GatingSet.

Usage

```
magic_template_predict(  
  gs_input,  
  parent_node,  
  channel_x,  
  channel_y,  
  ref_train,  
  ref_model_info,
```

```

    label_to_node,
    n_cores = 1,
    concavity_val = 50,
    add_to_gs = TRUE,
    return_gs_only = FALSE,
    ...
)

```

Arguments

<code>gs_input</code>	A GatingSet object.
<code>parent_node</code>	Character. Name of the parent node from which raw data are exported and under which predicted gates are added.
<code>channel_x</code>	Character. Name of the channel to use on the x-axis.
<code>channel_y</code>	Character. Name of the channel to use on the y-axis.
<code>ref_train</code>	Data frame. Training data generated by <code>get_train_data()</code> , typically <code>train_out\$ref_train</code> from <code>magic_template_train()</code> .
<code>ref_model_info</code>	List. Trained model information returned by <code>magicTrain()</code> , typically <code>train_out\$ref_model_info</code> from <code>magic_template_train()</code> .
<code>label_to_node</code>	Named character vector. Names are predicted labels and values are the corresponding GatingSet node names. For example, <code>c("1" = "Live_cells", "2" = "Dead_cells")</code> .
<code>n_cores</code>	Integer. Number of cores used for prediction and polygon extraction. Default is 1.
<code>concavity_val</code>	Numeric. Concavity parameter passed to <code>flowmagic_pred_to_poly_gates()</code> . Higher values generally produce smoother polygon gates. Default is 50.
<code>add_to_gs</code>	Logical. If TRUE, convert predicted labels to polygon gates and add or update them in the GatingSet. If FALSE, only prediction results are returned. Default is TRUE.
<code>return_gs_only</code>	Logical. If TRUE, return only the modified GatingSet. Default is FALSE.
<code>...</code>	Additional arguments passed to <code>export_raw_gs_plots()</code> .

Details

This function exports bivariate expression data from a selected GatingSet node, predicts event labels using a previously trained flowMagic model, converts selected predicted labels into polygon gates, and optionally adds or updates these gates in the input GatingSet.

The `label_to_node` argument controls how predicted labels are mapped to GatingSet node names. For example, `c("1" = "Live_cells", "2" = "Dead_cells")` converts predicted label "1" into a node named "Live_cells" and predicted label "2" into a node named "Dead_cells".

Value

If `return_gs_only = TRUE`, a modified GatingSet. Otherwise, a list with four elements: `gs`, the modified or original GatingSet; `list_test_data`, the exported test data; `list_dfs_pred`, the predicted labeled data frames; and `list_poly_gates`, the polygon gates generated for each requested node.

Examples

```

pred_out <- magic_template_predict(
  gs_input = gs,
  parent_node = "root",
  channel_x = "FSC-A",
  channel_y = "LIVE DEAD Blue-A",
  ref_train = train_out$ref_train,
  ref_model_info = train_out$ref_model_info,
  label_to_node = c("1" = "Live_cells", "2" = "Dead_cells"),
  n_cores = 1,
  add_to_gs = TRUE
)

gs <- pred_out$gs

```

magic_template_train *magic_template_train*

Description

Build a template training set from manually gated samples and train a flowMagic classification model.

Usage

```

magic_template_train(
  gs_input,
  parent_node,
  samples_id,
  channel_x,
  channel_y,
  size_points = 0.5,
  n_gates = 1,
  input_list_out_final = NULL,
  train_model = "rf",
  ...
)

```

Arguments

gs_input	A GatingSet object.
parent_node	Character. Name of the parent node from which expression data are extracted for template gating.
samples_id	Character vector. Sample names to manually gate as template samples.
channel_x	Character. Name of the channel to plot on the x-axis.
channel_y	Character. Name of the channel to plot on the y-axis.

size_points	Numeric. Point size used in the interactive gating plot. Default is 0.5.
n_gates	Integer. Number of template gates to draw per selected sample. Each gate is assigned a label from 1 to n_gates. Default is 1.
input_list_out_final	Optional list. Existing template data object, usually from a previous call to magic_template_train()\$list_out_final. If supplied, newly gated templates are added to this existing template set. Samples with matching names are replaced.
train_model	Character. Model type passed to magicTrain(). Default is "rf".
...	Additional arguments passed to magicGating().

Details

This function performs one or more rounds of interactive manual gating on selected samples. Each gate is assigned a different label. If more than one gate is drawn, the labeled outputs are merged into a single template data object. The resulting templates are converted into training data using get_train_data() and used to train a model with magicTrain().

Existing template data can be supplied through input_list_out_final, allowing users to add new manually gated templates without redrawing previous ones.

Value

A list with five elements: ref_model_info, the trained model information returned by magicTrain(); ref_train, the training data returned by get_train_data(); list_out_final, the complete merged template set; list_out_new, the newly generated template data; and all_list_out_obj, the raw outputs from each call to magicGating().

Examples

```
train_out <- magic_template_train(
  gs_input = gs,
  parent_node = "root",
  samples_id = c("sample_01.fcs", "sample_02.fcs"),
  channel_x = "FSC-A",
  channel_y = "LIVE DEAD Blue-A",
  n_gates = 2,
  train_model = "rf"
)

# Add additional template samples later
train_out2 <- magic_template_train(
  gs_input = gs,
  parent_node = "root",
  samples_id = c("sample_03.fcs"),
  channel_x = "FSC-A",
  channel_y = "LIVE DEAD Blue-A",
  n_gates = 2,
  input_list_out_final = train_out$list_out_final
)
```

map_to_parent	<i>map_to_parent</i>
---------------	----------------------

Description

function to get binary dataset with gate assignation based on the mother population of the selected pop (instead of the root).

Usage

```
map_to_parent(gh, binary_df)
```

Arguments

gh	GatingHierarchy.
binary_df	Dataframe generated by the map_to_root function.

Value

Dataframe.

Examples

```
map_to_parent()
```

map_to_root	<i>map_to_root</i>
-------------	--------------------

Description

it generates a dataset indicating what cells belong to the selected pop (1) 0 otherwise. Old name: pre_process_manual_binary() The dataset is always the dataset of the Root pop. gh and pop are mandatory arguments, dim and mode have a default value.

Usage

```
map_to_root(gh, pop)
```

Arguments

gh	GatingHierarchy.
pop	Name of the population to get events assignments.

Value

Dataframe.

Examples

```
map_to_root()
```

```
merge_magicGating_labels  
    merge_magicGating_labels
```

Description

function to merge list of gated data when gating multiple gates.

Usage

```
merge_magicGating_labels(list_out_1, list_out_2, gated_data_only = F)
```

Arguments

`list_out_1` List of Dataframes to update.
`list_out_2` List of Dataframes to merge with `list_out_1` dataframes.
`gated_data_only` Input contains only gated data,not polygons.Default to False.

Value

list of Dataframes

Examples

```
merge_magicGating_labels()
```

```
name_pop_gating      name_pop_gating
```

Description

function to get the name of all the pops of the gating hierarchy.

Usage

```
name_pop_gating(gh)
```

Arguments

`gh` GatingHierarchy.

Value

Vector of characters.

Examples

```
name_pop_gating()
```

post_process_gates	<i>post_process_gates</i>
--------------------	---------------------------

Description

function to post process the events after model prediction.

Usage

```
post_process_gates(  
  gated_df,  
  n_cores = 1,  
  thr_dist = 0.15,  
  include_zero = F,  
  remove_centroids = T,  
  type = "dist",  
  concavity_val = 5,  
  normalize_data = T,  
  ...  
)
```

Arguments

gated_df	dataframe with labels (third column).
n_cores	Number of cores. Default to 1.
thr_dist	Distance threshold for centroids calculation. Default to 0.15.
include_zero	Consider centroid of label 0. Default to False.
remove_centroids	Remove centroids too near each other based on thr_dist value.
type	Type of post-processing.
concavity_val	Concavity of polygons for the "polygon" type of post-processing

Value

Dataframe.

Examples

```
post_process_gates()
```

process_test_data	<i>process_test_data</i>
-------------------	--------------------------

Description

function to get test data correctly formatted.

Usage

```
process_test_data(  
  test_data,  
  prop_down = NULL,  
  n_points_per_plot = 500,  
  normalize_data = T  
)
```

Arguments

- test_data Dataframe of bivariate markers expression.
- prop_down Proportion of events (downsampling). Default to NULL (downsampling using number of points).
- n_points_per_plot Number of points for downsampling.
- normalize_data If True, data is normalized to 0-1 range. Default to True.

Value

Dataframe.

Examples

```
process_test_data()
```

range01	<i>range01</i>
---------	----------------

Description

function to put data in range 0-1.

Usage

```
range01(x)
```

Arguments

x Vector of numbers to scale.

Value

Vector of numbers.

Examples

range01()

smooth_hull	<i>smooth_hull</i>
-------------	--------------------

Description

function to smooth concave polygons

Usage

smooth_hull(hull_df, spar = 0.7)

Arguments

hull_df	Dataframe generate by concaveman functions inside get_huget_hull_all_gates function
spar	Spar value to regulate smoothing process: higher value (max 1) higher smoothing.
buffer_dist	Regulate buffer distance before smoothing. Default to 500.

Value

Dataframe.

Examples

smooth_hull()

```
update_label_association  
    update_label_association
```

Description

function to update label association.

Usage

```
update_label_association(df)
```

Arguments

df Dataframe.

Value

Vector.

Examples

```
update_label_association()
```

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